THE OSLO RICE PILE RELAXES EXACTLY IN $L(L+1)$ STEPS

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Abstract. Dhar stated that the evolution operator $\mathcal{W}$ of the Oslo rice pile on L sites satisfies $\mathcal{W}^{n+1} = \mathcal{W}^n$ with $n = L(L+1)/2$. This is false: the least such n is $L(L+1)$. We prove that a configuration σ with g grains reaches the steady state after exactly $L(L+1) - g + [\,\sigma_L = 2\,]$ added grains, and that Dhar's decorrelation time $L(L+1)/2$ is nevertheless exact on recurrent configurations.

1. Introduction

The Oslo rice pile [2, 7] is a one-dimensional sandpile in which the critical slope of a site is drawn afresh, equal to 1 with probability q and to 2 with probability $p = 1 - q$, each time the site topples. It is driven by dropping grains at the top of the pile and has been studied as a model of self-organized criticality [1, 11] and of interface depinning [9, 10, 8]. In [5] Dhar showed that, in slope variables, the model is an abelian distributed processors network [4], a stochastic relative of the abelian sandpile [3], wrote down the operators $a_1, \dots, a_L$ that add a particle at a site and relax, and drew strong conclusions about the evolution operator $\mathcal{W} = a_1$ acting on the 3^L-dimensional space $\mathcal{V}$ spanned by the stable configurations. The abstract of [5] reads, in part: "We show that the time evolution operator $\mathcal{W}$ for the system satisfies the equation $\mathcal{W}^{n+1} = \mathcal{W}^n$ where $n = L(L+1)/2$ for a pile with L sites."

This identity is false for every $L \geq 1$. For $L = 1$ it would say $\mathcal{W}^2 = \mathcal{W}$, but $\mathcal{W}|0\rangle = |1\rangle$ while $\mathcal{W}^2|0\rangle = p|2\rangle + q|1\rangle$. The body of [5] proves the weaker identity $\mathcal{W}^{L(L+1)+1} = \mathcal{W}^{L(L+1)}$ and then states, for every recurrent configuration $\mathcal{C}$ with m grains,

$$a_1^{L(L+1)-m}|\mathcal{C}\rangle = a_1^2 a_2^2 \cdots a_L^2 |0, \dots, 0\rangle, \tag{1}$$

from which it deduces that $\mathcal{W}^{L(L+1)/2}$ sends every recurrent configuration to the steady state $|\Psi\rangle$ and that every connected time-dependent correlation function vanishes beyond time difference $L(L+1)/2$. The configuration $\mathcal{C} = (2, \dots, 2)$ is recurrent and has $m = L(L+1)$ grains, so (1) would give $|2, \dots, 2\rangle = |\Psi\rangle$, which is false because Ψ also charges $(1, \dots, 1)$ (Proposition 6.1). The present paper determines the exact relaxation time of every configuration and thereby settles which of the statements of [5] hold.

Throughout, $L \geq 1$ and $0 < p < 1$. A stable configuration is a vector $\sigma = (\sigma_1, \dots, \sigma_L) \in \{0, 1, 2\}^L$ of slopes. Its number of grains is[1]

$$g(\sigma) = \sum_{x=1}^{L} x\, \sigma_x .$$

Let $r(\sigma)$ be the length of the longest block of 2's ending at site L, so that $r(\sigma) = 0$ when $\sigma_L \neq 2$, and write $\mathbf{2} = (2, \dots, 2)$, $\mathbf{1} = (1, \dots, 1)$ and $u = (1, 2, \dots, 2)$. The vector $v_u = a_1 a_2^2 \cdots a_L^2 |0\rangle$ is the law of the pile obtained from the empty pile by adding one particle at site 1 and two at each other site.

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[1] If h_x is the height of column x and $h_{L+1} = 0$, then $\sigma_x = h_x - h_{x+1}$ and $\sum_x h_x = \sum_x \sum_{y \geq x} \sigma_y = \sum_y y\, \sigma_y$.

Theorem A. *For every stable configuration σ and every $k \geq 0$,*

$$\mathcal{W}^k|\sigma\rangle = |\Psi\rangle \iff k \geq K(\sigma), \qquad K(\sigma) = L(L+1) - g(\sigma) + [\,\sigma_L = 2\,].$$

Moreover $K(\sigma) \geq 1$, and with $r = r(\sigma)$,

$$\mathcal{W}^{K(\sigma)-1}|\sigma\rangle = \begin{cases} v_u, & r = 0, \\ p^{-r}|\Psi\rangle + \big(1 - p^{-r}\big)v_u, & r \geq 1. \end{cases}$$

The second statement shows exactly what goes wrong one step before relaxation: when the last slope is 2, the steady state appears with weight $p^{-r} > 1$, compensated by the negative weight $1 - p^{-r}$ of a vector that has not yet reached the top configuration **2**. The recurrent configurations are those charged by Ψ; they are described in Section 6.

Corollary B.

(1) *The least n with $\mathcal{W}^{n+1} = \mathcal{W}^n$ on $\mathcal{V}$ is $L(L+1)$, and $|0, \dots, 0\rangle$ is the only configuration that needs $L(L+1)$ grains. The minimal polynomial of $\mathcal{W}$ is $x^{L(L+1)}(x-1)$, so $\mathcal{W}$ has a nilpotent Jordan block of size $L(L+1)$.*

(2) *On the span $\mathcal{V}_{\mathrm{rec}}$ of the recurrent configurations the least such n is $L(L+1)/2$, and the minimal polynomial of $\mathcal{W}|_{\mathcal{V}_{\mathrm{rec}}}$ is $x^{L(L+1)/2}(x-1)$.*

(3) *In the steady state every connected correlation $C_{XY}(\tau)$ vanishes for $\tau \geq L(L+1)/2$, and there are observables with $C_{XY}(L(L+1)/2 - 1) < 0$.*

(4) *Identity* (1) *holds for every configuration $\mathcal{C}$ with $\mathcal{C}_L \neq 2$, recurrent or not, and fails for every configuration with $\mathcal{C}_L = 2$; recurrent configurations with $\mathcal{C}_L = 2$ exist for every L.*

So the statement in the abstract of [5] is true on $\mathcal{V}_{\mathrm{rec}}$ and false on $\mathcal{V}$, the identity in its body is optimal, and its physical conclusions about correlations are correct and sharp; what fails is (1) and the operator identity as stated. The proofs are self-contained. Section 2 sets up the model and its abelian property, Section 3 reduces products of the operators a_x to a deterministic chip-firing game, Section 4 expands every configuration in the vectors $a^m|0\rangle$, Section 5 proves Theorem A, Section 6 describes the recurrent configurations, and Section 7 proves Corollary B. Figure 1 shows $K(\sigma)$ for all configurations at $L = 5$.

2. The model and its abelian property

2.1. **Particles.** Following [5, Sections 3–4] we describe the pile by its slopes and call the units of slope particles. Adding a grain at the top raises σ_1 by one, that is, it adds a particle at site 1. When a particle arrives at a site x in state σ_x:

- if $\sigma_x = 0$, then σ_x becomes 1;
- if $\sigma_x = 1$, then with probability p the slope becomes 2, and with probability q the site topples: σ_x becomes 0 and one particle is sent to each of $x - 1$ and $x + 1$;
- if $\sigma_x = 2$, the site topples: σ_x becomes 1 and one particle is sent to each of $x - 1$ and $x + 1$.

A particle sent to site 0 leaves the system, and a particle sent to site $L + 1$ is delivered to site L.[2] This is the rule by which Dhar replaces the critical thresholds [5, Section 3].[3] The operators of [5], and hence all its statements, refer to this process.

To make the randomness explicit, give every site x an infinite sequence $\xi_x = (\xi_x(1), \xi_x(2), \dots)$ of symbols in $\{\mathsf{S}, \mathsf{T}\}$ (stay, topple). When a particle arrives at x while $\sigma_x = 1$, the site reads

[2] A toppling at $x < L$ moves one grain from column x to column $x+1$. A toppling at L, followed by the arrival of the particle sent back to L, changes (σ_{L-1}, σ_L) by $(+1, -1)$, which is rule (ii)c of [5]: one grain leaves the pile.

[3] In the threshold description, from the toppling of x that draws a threshold until the next arrival at slope 1, the slope of x stays at most 1, so the threshold has not been tested and is still distributed as a fresh draw at that arrival; a site at slope 2 has threshold 2. Processing one particle at a time, the two descriptions therefore give the same process.

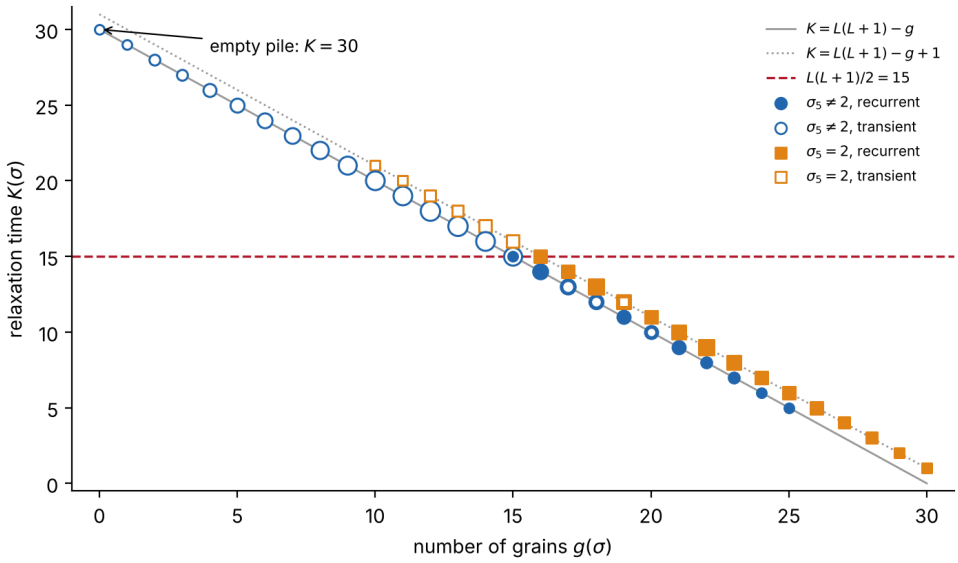


FIGURE 1. The relaxation time $K(\sigma)$ against the number of grains $g(\sigma)$ for all 3^5 configurations at $L = 5$. Configurations with $\sigma_5 = 2$ lie one step above the line $K = 30 - g$. Filled marks are recurrent; all of them lie on or below the dashed level $L(L+1)/2 = 15$, which is attained. The empty pile alone needs $L(L+1) = 30$ grains, whereas the abstract of [5] asserts 15 for every configuration. Mark size grows with the number of configurations at a point; a filled mark inside an open one means that both recurrent and transient configurations occur there.

the next unread symbol of ξ_x and stays at 2 on S or topples on T. When the sequences are independent with independent entries, each equal to S with probability p, this reproduces the rule above.

A *state* consists of a configuration $\sigma \in \mathbb{Z}_{\geq 0}^L$ (not necessarily stable during relaxation), a vector $\eta \in \mathbb{Z}_{\geq 0}^L$ of pending particles, and the number of symbols read at each site. A *step* chooses a site x with $\eta_x \geq 1$ and processes one of its pending particles. Starting from a stable σ and an input $\lambda \in \mathbb{Z}_{\geq 0}^L$, with $\eta = \lambda$ initially, a *relaxation* is a sequence of steps; it is *complete* if it ends with $\eta = 0$. The relaxation of a single particle at a stable site always ends in a stable slope, so a complete relaxation ends in a stable configuration.

Throughout we use e_x for the unit vector at x, together with the conventions

$$e_0 = 0, \qquad e_{L+1} = e_L,$$

and the $L \times L$ matrix D with columns $De_x = 2e_x - e_{x-1} - e_{x+1}$; explicitly D is tridiagonal with diagonal $(2, \ldots, 2, 1)$ and off-diagonal entries -1.[4] Its inverse is

$$D^{-1} = \big(\min(x, y)\big)_{x,y=1}^{L}, \tag{2}$$

as one checks column by column: for $x < L$, $2\min(x, y) - \min(x-1, y) - \min(x+1, y) = \delta_{xy}$, and $\min(L, y) - \min(L-1, y) = \delta_{Ly}$. In particular D^{-1} has nonnegative entries. The *weight* of a vector $m \in \mathbb{Z}^L$ is $w(m) = \sum_x x\, m_x$; thus $g(\sigma) = w(\sigma)$. Since $\sum_x x\, D_{xy} = 0$ for $y < L$ and $= 1$ for $y = L$,

$$w(m - Df) = w(m) - f_L \qquad (m, f \in \mathbb{Z}^L). \tag{3}$$

[4]For $x = L$ the convention gives $De_L = e_L - e_{L-1}$.

2.2. **The effective configuration.** Along a relaxation put $\tilde{\sigma} = \sigma + \eta$. A step that does not topple moves one particle from η_x to σ_x and leaves $\tilde{\sigma}$ unchanged. A toppling at x lowers $\sigma_x + \eta_x$ by 2 and sends one particle to each of $x \pm 1$ under the conventions above, so it replaces $\tilde{\sigma}$ by $\tilde{\sigma} - De_x$. A toppling at x requires $\sigma_x \geq 1$ and $\eta_x \geq 1$, so $\tilde{\sigma}_x \geq 2$; if it happens at $\sigma_x = 2$ then $\tilde{\sigma}_x \geq 3$.

Lemma 2.1. *Let σ be stable and $\lambda \in \mathbb{Z}^L_{\geq 0}$. Every relaxation from (σ, λ) has at most*

$$M(\sigma, \lambda) = |\lambda| + 2\sum_{x=1}^{L} \big(D^{-1}(\sigma + \lambda)\big)_x$$

steps, where $|\lambda| = \sum_x \lambda_x$. In particular every relaxation can be continued to a complete one.

Proof. Let $N \in \mathbb{Z}^L_{\geq 0}$ count the topplings so far. Then $\tilde{\sigma} = \sigma + \lambda - DN \geq 0$, so $DN \leq \sigma + \lambda$ and, D^{-1} being nonnegative, $N \leq D^{-1}(\sigma + \lambda)$. Each step processes a particle that was either in the input or sent by one of the topplings, and each toppling sends at most two particles. Hence the number of steps is at most $|\lambda| + 2\sum_x N_x \leq M(\sigma, \lambda)$. As long as $\eta \neq 0$ a further step is possible, so the bound forces every relaxation to reach $\eta = 0$ if continued. □

Lemma 2.2. *Fix the sequences ξ, a stable σ and an input λ. All complete relaxations from (σ, λ) process the same number of particles at each site and end in the same state.*

Proof. Given σ_x and ξ_x, the behaviour of site x depends only on the number n_x of particles it has processed: its slope, the number of symbols it has read and the number $c_x(n_x)$ of its topplings are functions of n_x, and c_x is nondecreasing. Put $c_0 = c_{L+1} = 0$. When $n \in \mathbb{Z}^L_{\geq 0}$ particles have been processed, the number of pending particles at x is

$$\eta_x(n) = \lambda_x + c_{x-1}(n_{x-1}) + c_{x+1}(n_{x+1}) + [\,x = L\,]\, c_L(n_L) - n_x.$$

For $y \neq x$ this is nondecreasing in n_y. Let α and β be complete relaxations with final counts n^α and n^β, so $\eta(n^\beta) = 0$. Suppose some step of α processes a particle at a site x that has already processed n^β_x particles in α, and take the first such step. Just before it the counts n satisfy $n \leq n^\beta$ and $n_x = n^\beta_x$, hence $\eta_x(n) \leq \eta_x(n^\beta) = 0$, and the step is impossible. Thus $n^\alpha \leq n^\beta$, and by symmetry $n^\alpha = n^\beta$. The final state is a function of the counts. □

2.3. **The operators.** For a stable σ and an input λ let $\Phi(\sigma, \lambda; \xi)$ be the final configuration of any complete relaxation, and let P_λ be the linear operator on $\mathcal{V}$ with

$$P_\lambda |\sigma\rangle = \sum_{\sigma'} \Pr\big(\Phi(\sigma, \lambda; \xi) = \sigma'\big) |\sigma'\rangle,$$

the sequences ξ being random as above. Only finitely many symbols are read (Lemma 2.1), so these probabilities are well defined. Put $a_x = P_{e_x}$ for $1 \leq x \leq L$, $a_0 = 1$ and $a_{L+1} = a_L$. Then $\mathcal{W} = a_1$ is the transition matrix of the driven pile, and $\mathcal{W}$ maps probability vectors to probability vectors.

Proposition 2.3. *For inputs λ, λ' we have $P_{\lambda+\lambda'} = P_{\lambda'} P_\lambda$. Consequently the a_x commute and $P_\lambda = \prod_x a_x^{\lambda_x}$.*

Proof. Relax $(\sigma, \lambda + \lambda')$ in two phases. In the first, process only the particles of λ and those they generate, exactly as in a complete relaxation of (σ, λ); this is a legitimate sequence of steps because the extra pending particles λ' only enlarge η. Then complete the relaxation. By Lemma 2.2 the result is $\Phi(\sigma, \lambda + \lambda'; \xi)$. Run the first phase by a fixed rule (always the smallest site with a pending particle of the first phase). Whether it ends in the configuration σ'' after reading k_x symbols at each x is decided by the first k_x symbols of each ξ_x. Conditionally on

this event, the unread parts $(\xi_x(k_x+1), \xi_x(k_x+2), \dots)$ are again independent with independent entries of the same law, and the second phase is a complete relaxation of (σ'', λ') driven by them. Summing over the events gives $P_{\lambda+\lambda'}|\sigma\rangle = P_{\lambda'}P_\lambda|\sigma\rangle$. Since $\lambda+\lambda' = \lambda'+\lambda$, the operators P_λ commute, and induction on $|\lambda|$ gives $P_\lambda = \prod_x a_x^{\lambda_x}$. □

Proposition 2.4 (Dhar's relations)**.** *Let σ be stable and $1 \le x \le L$. Then*

(R0) $a_x|\sigma\rangle = |\sigma + e_x\rangle$ *if* $\sigma_x = 0$,

(R1) $a_x|\sigma\rangle = p|\sigma + e_x\rangle + q\, a_{x-1}a_{x+1}|\sigma - e_x\rangle$ *if* $\sigma_x = 1$,

(R2) $a_x|\sigma\rangle = a_{x-1}a_{x+1}|\sigma - e_x\rangle$ *if* $\sigma_x = 2$.

Proof. Process the added particle first. If $\sigma_x = 0$ the configuration $\sigma + e_x$ is stable. If $\sigma_x = 1$ the first symbol of ξ_x is S with probability p, giving the stable $\sigma + e_x$; otherwise the site falls to 0 and sends particles to $x \pm 1$. If $\sigma_x = 2$ it falls to 1 and sends particles to $x \pm 1$. In the toppling cases the remaining relaxation is a complete relaxation of $\sigma - e_x$ with input $e_{x-1} + e_{x+1}$, driven by the unread symbols, which are independent of the first one; by the argument of Proposition 2.3 its law is $P_{e_{x-1}+e_{x+1}}|\sigma - e_x\rangle = a_{x-1}a_{x+1}|\sigma - e_x\rangle$, with $e_0 = 0$, $e_{L+1} = e_L$. □

3. Monomials and chip-firing

For $m \in \mathbb{Z}^L_{\ge 0}$ write $a^m = a_1^{m_1} \cdots a_L^{m_L}$ and $v_m = a^m|0\rangle$, where $|0\rangle = |0, \dots, 0\rangle$. By Proposition 2.3, $v_m = P_m|0\rangle$ is a probability vector.

Lemma 3.1. $a_x^3 = a_{x-1}a_x a_{x+1}$ *for* $1 \le x \le L$.

Proof. We check the identity on each basis vector, using commutativity and (R0)–(R2). Write $|j\rangle$ for $|\sigma\rangle$ with $\sigma_x = j$ and the other slopes fixed, and $b = a_{x-1}a_{x+1}$. First, $a_x^2|1\rangle = a_x(p|2\rangle + qb|0\rangle) = pb|1\rangle + qb\, a_x|0\rangle = b|1\rangle$. Hence

$$a_x^3|0\rangle = a_x^2|1\rangle = b|1\rangle = b\, a_x|0\rangle, \qquad a_x^3|1\rangle = a_x b|1\rangle = b\, a_x|1\rangle,$$
$$a_x^3|2\rangle = a_x^2 b|1\rangle = b\, a_x^2|1\rangle = b\, b|1\rangle = b\, a_x|2\rangle.$$ □

Call x *unstable* in $m \in \mathbb{Z}^L_{\ge 0}$ if $m_x \ge 3$; *firing* x replaces m by $m - De_x$, that is, m_x drops by 2 and each of $x \pm 1$ gains 1 (so m_L drops by 1 when $x = L$, and nothing is added at site 0). A firing is *legal* if x is unstable. A vector is *stable* if $m \le \mathbf{2}$ entrywise, and these are exactly the stable configurations. By Lemma 3.1, if $m_x \ge 3$ then

$$a^m = a^{m-3e_x}a_x^3 = a^{m-3e_x}a_{x-1}a_x a_{x+1} = a^{m-De_x}. \tag{4}$$

Lemma 3.2 (least action)**.** *Let $m \in \mathbb{Z}^L_{\ge 0}$ and $f \in \mathbb{Z}^L_{\ge 0}$ with $(m - Df)_x \le 2$ for every x. Then every sequence of legal firings starting at m fires each x at most f_x times.*

Proof. Suppose not, and consider the first firing in the sequence that fires some x for the (f_x+1)-st time. Just before it the firing counts F satisfy $F \le f$ and $F_x = f_x$. The off-diagonal entries of D are ≤ 0, so

$$(m - DF)_x = m_x - D_{xx}f_x - \sum_{y \ne x} D_{xy}F_y \le m_x - D_{xx}f_x - \sum_{y \ne x} D_{xy}f_y = (m - Df)_x \le 2,$$

and x is not unstable, a contradiction. □

This principle is due to Fey, Levine and Peres [6] for the abelian sandpile; the proof above is theirs, adapted to the matrix D.

Corollary 3.3. *Every sequence of legal firings from m is finite, and all maximal sequences have the same firing vector $F(m)$ and end in the same stable vector* $\mathrm{stab}(m) = m - DF(m)$. *Moreover:*

(1) $a^m = a^{\mathrm{stab}(m)}$ *and* $v_m = v_{\mathrm{stab}(m)}$*;*
(2) $w(\mathrm{stab}(m)) = w(m) - F_L(m)$*;*
(3) $\mathrm{stab}(m+m') = \mathrm{stab}(\mathrm{stab}(m)+m')$ *for* $m' \in \mathbb{Z}^L_{\geq 0}$.

Proof. Along legal firings the vector stays nonnegative, so the firing vector F satisfies $DF \leq m$ and hence $F \leq D^{-1}m$ by (2); every sequence is therefore finite. If α, β are maximal, the end vector $m - DF^\beta$ of β is stable, so Lemma 3.2 with $f = F^\beta$ gives $F^\alpha \leq F^\beta$, and symmetry gives equality. Item (1) follows from (4), item (2) from (3). For (3), a maximal sequence for m remains legal from $m + m'$, since adding chips never makes a firing illegal; continuing it maximally stabilizes $\mathrm{stab}(m) + m'$, and the result is $\mathrm{stab}(m + m')$ by uniqueness. □

The stable vectors have weight at most $w(\mathbf{2}) = L(L+1)$, with equality only for $\mathbf{2}$, and $\mathbf{2} - e_1 = u$ is the only stable vector of weight $L(L+1) - 1$.[5]

Lemma 3.4. *Let s be stable. If $s \neq \mathbf{2}$, then* $\mathrm{stab}(s + e_1)$ *has weight $w(s) + 1$ and site L does not fire. If $s = \mathbf{2}$, then* $\mathrm{stab}(\mathbf{2} + e_1) = \mathbf{2}$.

Proof. If $s \neq \mathbf{2}$, let j be the least site with $s_j \leq 1$. If $j = 1$ nothing fires. Otherwise fire $1, 2, \ldots, j-1$ in this order. Before the firing of $i \leq j-1$ the vector is $(2, \ldots, 2, 1, 3, 2, \ldots)$ with the 3 at position i (for $i = 1$, simply 3 at position 1), so the firing is legal; after it the vector is $(2, \ldots, 2, 1, 3, \ldots)$ with the 1 at position i and the 3 moved to $i+1$, except that after firing $j-1$ the entry at j is $s_j + 1 \leq 2$. The final vector is stable, no firing occurred at L because $j - 1 < L$, and by Corollary 3.3(2) the weight rose by 1. If $s = \mathbf{2}$, the same sequence fires $1, \ldots, L-1$ and produces $(2, \ldots, 2, 1, 3)$; firing L then gives $\mathbf{2}$. (For $L = 1$, the vector 3 fires once to 2.) □

Corollary 3.5. *Let s be stable and $k \geq 0$. If $k \geq L(L+1) - w(s)$ then* $\mathrm{stab}(s + ke_1) = \mathbf{2}$. *If $k < L(L+1) - w(s)$ then* $\mathrm{stab}(s + ke_1)$ *is stable of weight $w(s) + k$. In particular* $\mathrm{stab}(L(L+1)e_1) = \mathbf{2}$.

Proof. By Corollary 3.3(3), $\mathrm{stab}(s + (k+1)e_1) = \mathrm{stab}(\mathrm{stab}(s + ke_1) + e_1)$. By Lemma 3.4 each added chip raises the weight by exactly one until the weight $L(L+1)$, that is the vector $\mathbf{2}$, is reached, and $\mathbf{2}$ is then fixed. □

4. Expansion of a configuration and the steady state

Lemma 4.1. *Let σ be stable and $T = \{x : \sigma_x = 2\}$, $t = |T|$. For $A \subseteq T$ put $\tau_A = \sigma - D\mathbf{1}_A$, where $\mathbf{1}_A = \sum_{x \in A} e_x$. Then $\tau_A \geq 0$ and*

$$|\sigma\rangle = p^{-t} \sum_{A \subseteq T} (-q)^{|A|}\, v_{\tau_A}.$$

Proof. By (R0), $a_x|\rho\rangle = |\rho + e_x\rangle$ when $\rho_x = 0$; and by (R0)–(R1), $a_x^2|\rho\rangle = a_x|\rho + e_x\rangle = p|\rho + 2e_x\rangle + q\, a_{x-1}a_{x+1}|\rho\rangle$ when $\rho_x = 0$, that is

$$|\rho + 2e_x\rangle = p^{-1}\big(a_x^2 - q\, a_{x-1}a_{x+1}\big)|\rho\rangle \qquad (\rho \text{ stable},\ \rho_x = 0). \tag{5}$$

Let σ° be σ with every 2 replaced by 0; then $|\sigma^\circ\rangle = a^{\sigma^\circ}|0\rangle$ by repeated use of (R0). Applying (5) once for each $x \in T$ and using commutativity,

$$|\sigma\rangle = p^{-t} \prod_{x \in T} \big(a_x^2 - q\, a_{x-1}a_{x+1}\big) a^{\sigma^\circ}|0\rangle = p^{-t} \sum_{A \subseteq T} (-q)^{|A|} a^{\tau_A}|0\rangle,$$

because the monomial chosen from the factors is $a^{\sigma^\circ} \prod_{x \in T \setminus A} a_x^2 \prod_{x \in A} a_{x-1}a_{x+1}$, whose exponent is $\sigma^\circ + 2\mathbf{1}_{T \setminus A} + \sum_{x \in A}(e_{x-1} + e_{x+1}) = \sigma - D\mathbf{1}_A$. □

[5] If $s \leq \mathbf{2}$ then $w(\mathbf{2}) - w(s) = \sum_x x(2 - s_x)$ is a sum of nonnegative terms, which equals 1 only for $2 - s = e_1$.

By Corollary 3.3(2) and (3), $w(\tau_A) = g(\sigma) - [\,L \in A\,]$, and

$$s_A := \mathrm{stab}(\tau_A) \quad \text{has weight} \quad w(s_A) = g(\sigma) - \mu_A, \qquad \mu_A := [\,L \in A\,] + F_L(\tau_A). \tag{6}$$

Define the steady state by $|\Psi\rangle = v_{\mathbf{2}} = a_1^2 a_2^2 \cdots a_L^2 |0\rangle$.

Lemma 4.2. *$|\Psi\rangle = \mathcal{W}^{L(L+1)}|0\rangle = a_1|\mathbf{2}\rangle$ and $\mathcal{W}|\Psi\rangle = |\Psi\rangle$.*

Proof. By Corollaries 3.3(1) and 3.5, $\mathcal{W}^{L(L+1)}|0\rangle = v_{L(L+1)e_1} = v_{\mathbf{2}}$ and $\mathcal{W}|\Psi\rangle = v_{\mathbf{2}+e_1} = v_{\mathbf{2}}$. For the remaining identity, if $\rho_x = 2$ then by (R2), commutativity and (R0)–(R1),

$$a_x^3|\rho - 2e_x\rangle = a_x^2|\rho - e_x\rangle = a_x\big(p|\rho\rangle + q\, a_{x-1}a_{x+1}|\rho - 2e_x\rangle\big) = p\, a_x|\rho\rangle + q\, a_{x-1}a_{x+1}|\rho - e_x\rangle = a_x|\rho\rangle.$$

Let $\rho^{(j)} = (0, \ldots, 0, 2, \ldots, 2)$ with $j-1$ zeros. We claim $a_1|\mathbf{2}\rangle = a_1^2 \cdots a_{j-2}^2\, a_{j-1}a_j|\rho^{(j)}\rangle$ for $1 \le j \le L+1$ (for $j = 1$ the right side is $a_0 a_1|\mathbf{2}\rangle$). Given the claim for $j \le L$, the displayed identity and Lemma 3.1 give $a_j|\rho^{(j)}\rangle = a_j^3|\rho^{(j+1)}\rangle = a_{j-1}a_j a_{j+1}|\rho^{(j+1)}\rangle$, which is the claim for $j+1$. For $j = L+1$ the claim reads $a_1|\mathbf{2}\rangle = a_1^2 \cdots a_{L-1}^2 a_L a_{L+1}|0\rangle = v_{\mathbf{2}}$. □

Thus Ψ is the state $|\Psi_{st}\rangle$ of [5]. Two elementary facts about supports will replace any appeal to linear independence.

Lemma 4.3. *Let σ be stable and λ an input.*

1. *Every configuration in the support of $P_\lambda|\sigma\rangle$ has weight at most $g(\sigma) + w(\lambda)$.*
2. *$P_\lambda|\sigma\rangle$ charges $\mathrm{stab}(\sigma + \lambda)$ with positive probability.*

In particular $\Psi(\mathbf{2}) > 0$, while $v_u(\mathbf{2}) = 0$ and $(\mathcal{W}^k|\sigma\rangle)(\mathbf{2}) = 0$ whenever $g(\sigma) + k < L(L+1)$.

Proof. (1) Along a relaxation $\tilde\sigma$ starts at $\sigma + \lambda$, is unchanged by steps without toppling, and changes by $-De_x$ at topplings, which does not raise the weight by (3); at the end $\tilde\sigma$ is the final configuration. (2) Consider the event that the first $M(\sigma, \lambda)$ symbols of every ξ_x are S; it has probability $p^{LM(\sigma,\lambda)} > 0$, and by Lemma 2.1 no other symbol is read. On this event a site topples only on an arrival at slope 2, hence only when $\tilde\sigma_x \ge 3$, so the topplings form a sequence of legal firings for $\tilde\sigma$ that ends in a stable vector; by Corollary 3.3 the final configuration is $\mathrm{stab}(\sigma + \lambda)$. The last sentence follows from (1), from (2) with $\Psi = P_{L(L+1)e_1}|0\rangle$ and $\mathrm{stab}(L(L+1)e_1) = \mathbf{2}$, and from $w(u) = L(L+1) - 1$. □

5. Proof of Theorem A

Let σ be stable, with T, t, τ_A, s_A and μ_A as in Section 4, and let $R = \{L - r + 1, \ldots, L\}$ be the final block of 2's, $r = r(\sigma)$ (empty if $r = 0$).

Lemma 5.1. *For every $A \subseteq T$, $\mu_A = [\,A \cap R \neq \emptyset\,]$.*

Proof. Upper bound when $A \cap R = \emptyset$. Put $B = \{1, \ldots, L-r-1\}$ (empty if $r \ge L-1$). Site $L-r$, when it exists, is not in T, so $A \subseteq T \setminus R \subseteq B$. We apply Lemma 3.2 to $m = \tau_A$ and $f = \mathbf{1}_B - \mathbf{1}_A \ge 0$, for which $m - Df = \sigma - D\mathbf{1}_B$. For $x \in B$, $(\sigma - D\mathbf{1}_B)_x = \sigma_x - 2 + \#\{\text{neighbours of } x \text{ in } B\} \le 2$. A site $x \notin B$ has at most one neighbour in B, namely $x-1$ when $x = L-r$; there $\sigma_{L-r} \le 1$, so $(\sigma - D\mathbf{1}_B)_{L-r} \le 2$, and every other $x \notin B$ has $(\sigma - D\mathbf{1}_B)_x = \sigma_x \le 2$.[6] Hence $F_L(\tau_A) \le f_L = 0$. Also $L \notin A$, since $L \in R$ when $r \ge 1$ and $L \notin T$ when $r = 0$. So $\mu_A = 0$.

Upper bound in general. Lemma 3.2 with $f = \mathbf{1}_{\{1,\ldots,L\}} - \mathbf{1}_A$ applies, because $\sigma - D\mathbf{1}_{\{1,\ldots,L\}}$ has entries $\sigma_1 - 1$, σ_x for $1 < x < L$, and $\sigma_L - 1 + 1$ at L (for $L = 1$ the single entry is $\sigma_1 - 1$), all ≤ 2. Hence $F_L(\tau_A) \le 1 - [\,L \in A\,]$ and $\mu_A \le 1$.

Lower bound when $A \cap R \neq \emptyset$. If $L \in A$ then $\mu_A \ge 1$. Otherwise let $j = \max(A \cap R) < L$. The sites $j+1, \ldots, L$ lie in $R \setminus A$, so in τ_A they carry 2, except that $j+1$ receives one more

[6] D is symmetric and $D_{xy} = -1$ exactly when $|x - y| = 1$; so the entry at $x \notin B$ is σ_x plus the number of neighbours of x in B. When $r = 0$ the site $L - r$ is L, and the bound is $\sigma_L + 1 \le 2$.

chip from $j \in A$: τ_A reads $3, 2, \dots, 2$ on $j+1, \dots, L$. Firing $j+1, j+2, \dots, L$ in this order is legal: before the firing of i the entry at i is 3, and the firing turns the entry at $i+1$ from 2 into 3. This sequence contains a firing of L, and it extends to a maximal one; by Corollary 3.3, $F_L(\tau_A) \geq 1$. □

Proof of Theorem A. By Lemma 4.1, Proposition 2.3 and Corollary 3.3,

$$\mathcal{W}^k|\sigma\rangle = p^{-t} \sum_{A \subseteq T} (-q)^{|A|}\, v_{\mathrm{stab}(s_A + ke_1)}. \tag{7}$$

By (6) and Lemma 5.1, $w(s_A) = g(\sigma) - [\, A \cap R \neq \emptyset\,]$. If $r = 0$ then every s_A has weight $g(\sigma)$; if $r \geq 1$ then s_A has weight $g(\sigma)$ or $g(\sigma) - 1$ according as $A \cap R$ is empty or not. In both cases the largest value of $L(L+1) - w(s_A)$ is $K(\sigma)$. Note $K(\sigma) \geq 1$: if $g(\sigma) = L(L+1)$ then $\sigma = \mathbf{2}$ and $K = 1$, and otherwise $g(\sigma) < L(L+1)$.

If $k \geq K(\sigma)$, Corollary 3.5 gives $\mathrm{stab}(s_A + ke_1) = \mathbf{2}$ for all A, and (7) becomes

$$p^{-t} \sum_{A \subseteq T} (-q)^{|A|} |\Psi\rangle = p^{-t}(1-q)^t |\Psi\rangle = |\Psi\rangle.$$

Now let $k = K(\sigma) - 1$. If $r = 0$, every s_A has weight $g(\sigma) = L(L+1) - 1 - k$, so by Corollary 3.5 $\mathrm{stab}(s_A + ke_1)$ is the stable vector of weight $L(L+1) - 1$, which is u, and $\mathcal{W}^k|\sigma\rangle = v_u$. If $r \geq 1$, then $k = L(L+1) - g(\sigma)$; the s_A with $A \cap R = \emptyset$ reach $\mathbf{2}$, and those with $A \cap R \neq \emptyset$ stop at weight $L(L+1) - 1$, at u. Since

$$\sum_{A \subseteq T,\ A \cap R = \emptyset} (-q)^{|A|} = (1-q)^{t-r} = p^{t-r},$$

we get $\mathcal{W}^k|\sigma\rangle = p^{-t}\big(p^{t-r}|\Psi\rangle + (p^t - p^{t-r})v_u\big) = p^{-r}|\Psi\rangle + (1 - p^{-r})v_u$.

In both cases $\mathcal{W}^{K-1}|\sigma\rangle \neq |\Psi\rangle$: by Lemma 4.3 its value at $\mathbf{2}$ is 0 when $r = 0$ and $p^{-r}\Psi(\mathbf{2}) \neq \Psi(\mathbf{2})$ when $r \geq 1$, since $\Psi(\mathbf{2}) > 0$ and $p < 1$. Finally, if $\mathcal{W}^k|\sigma\rangle = |\Psi\rangle$ for some k, then $\mathcal{W}^{k'}|\sigma\rangle = \mathcal{W}^{k'-k}|\Psi\rangle = |\Psi\rangle$ for all $k' \geq k$ by Lemma 4.2; so $\mathcal{W}^k|\sigma\rangle \neq |\Psi\rangle$ for every $k < K(\sigma)$. □

Example 5.2. For $L = 2$, K takes the values $6, 5, 4, 4, 3, 2, 3, 2, 1$ on $00, 10, 20, 01, 11, 21, 02, 12, 22$ (here $\sigma_1\sigma_2$). The recurrent configuration $(0, 2)$ has $m = 4$ grains, and (1) would give $a_1^2|0,2\rangle = |\Psi\rangle$. In fact $a_1^2|0,2\rangle = p^{-1}|\Psi\rangle + (1-p^{-1})\, a_1 a_2^2|0\rangle$, whose value at $(2,2)$ is $p^{-1}\Psi(2,2) = p$, while $\Psi(2,2) = p^2$ by the computation of $a_1|2,2\rangle$ in [5, Section 4]; three grains are needed. For $L = 1$ the theorem gives $|2\rangle = p^{-1}|\Psi\rangle + (1-p^{-1})|1\rangle$, i.e. $|\Psi\rangle = p|2\rangle + q|1\rangle$.

6. Recurrent configurations

A configuration is *recurrent* if Ψ charges it. Since $\mathcal{W}|\Psi\rangle = |\Psi\rangle$, the support of Ψ is closed under the transitions of $\mathcal{W}$: if $\Psi(\rho) > 0$ and $\mathcal{W}$ moves ρ to ρ' with positive probability, then $\Psi(\rho') \geq \Psi(\rho)\, \mathcal{W}(\rho \to \rho') > 0$. Let $\mathcal{R}$ be the set of stable σ such that every 0 is followed by a 2 with only 1's in between: for every x with $\sigma_x = 0$ there is $j > x$ with $\sigma_j = 2$ and $\sigma_y = 1$ for $x < y < j$.

Proposition 6.1. *Every recurrent configuration lies in $\mathcal{R}$, and $\mathbf{1}$ is recurrent.*

Proof. Say that $\tilde\sigma \in \mathbb{Z}^L_{\geq 0}$ has property $(*)$ if for every x with $\tilde\sigma_x = 0$ there is $j > x$ with $\tilde\sigma_j \geq 2$ and $\tilde\sigma_y = 1$ for $x < y < j$. A stable vector has $(*)$ if and only if it lies in $\mathcal{R}$.

(a) If $\sigma \in \mathcal{R}$ then $\sigma + e_1$ has $(*)$: its zeros are zeros of σ other than site 1, and the condition at x only involves sites to the right of x.

(b) If $\tilde\sigma$ has $(*)$ and $\tilde\sigma_y \geq 2$, then $\tilde\sigma' = \tilde\sigma - De_y$ has $(*)$. Indeed, $\tilde\sigma'$ differs from $\tilde\sigma$ only at y, where it decreases, and at $y \pm 1$, where it increases by one (within $\{1, \dots, L\}$). Let $\tilde\sigma'_x = 0$. If $x > y$, then $x \geq y + 2$ because $\tilde\sigma'_{y+1} \geq 1$, and $\tilde\sigma'$ agrees with $\tilde\sigma$ from x on. If $x = y$, then $y < L$

and $\tilde\sigma_y = 2$; either $\tilde\sigma_{y+1} \geq 1$ and $j = y+1$ works, or $\tilde\sigma_{y+1} = 0$, $\tilde\sigma'_{y+1} = 1$, and the site j provided by $(*)$ for $\tilde\sigma$ at $y+1$ works, because $\tilde\sigma' = \tilde\sigma$ beyond $y+1$. If $x < y$, then $x \leq y-2$ because $\tilde\sigma'_{y-1} \geq 1$, so $\tilde\sigma_x = 0$; let j be the site given by $(*)$ for $\tilde\sigma$. As $\tilde\sigma_y \geq 2$ and $\tilde\sigma = 1$ on (x, j), we have $j \leq y$. If $j \leq y-2$ nothing changed on $[x, j]$; if $j = y-1$ then $\tilde\sigma'_j \geq 3$; if $j = y$ then $\tilde\sigma_{y-1} = 1$, so $\tilde\sigma'_{y-1} = 2$ and $j' = y-1$ works.

(c) Let $\sigma \in \mathcal{R}$. Along any relaxation of (σ, e_1) the vector $\tilde\sigma$ starts with $(*)$ by (a), is unchanged by steps without toppling, and changes by $-De_x$ with $\tilde\sigma_x \geq 2$ at topplings, so it keeps $(*)$ by (b). The final configuration is the final $\tilde\sigma$, hence lies in $\mathcal{R}$. Thus $\mathcal{W}$ maps $\mathcal{R}$ into combinations of elements of $\mathcal{R}$. As $\mathbf{2} \in \mathcal{R}$ and $|\Psi\rangle = \mathcal{W}|\mathbf{2}\rangle$ (Lemma 4.2), every recurrent configuration lies in $\mathcal{R}$.

(d) Consider the relaxation of $(\mathbf{2}, e_1)$ on the event that the first $M(\mathbf{2}, e_1)$ symbols of every ξ_x are T, which has positive probability. By Lemma 2.2 we may process the particles in any order. Site 1 receives the added particle at slope 2 and topples, sending a particle to site 2; processing that particle next, site 2 topples in turn, and so on up to site L. So every site receives a particle. A site at slope 2 falls to 1 when it receives one, and afterwards, on this event, an arrival at slope 1 produces 0 and an arrival at 0 produces 1. So the final configuration lies in $\{0,1\}^L$, and by (c) in $\mathcal{R}$. The only element of $\mathcal{R} \cap \{0,1\}^L$ is $\mathbf{1}$. Hence $\mathcal{W}(\mathbf{2} \to \mathbf{1}) > 0$, and as $\mathbf{2}$ is recurrent by Lemma 4.3, so is $\mathbf{1}$. □

Lemma 6.2. *Every $\sigma \in \mathcal{R}$ satisfies $g(\sigma) \geq L(L+1)/2 + [\,\sigma_L = 2\,]$.*

Proof. Map each x with $\sigma_x = 0$ to the site $j(x) > x$ of the definition of $\mathcal{R}$. The map is injective: if $x < x'$ had $j(x) = j(x')$, then x' would lie strictly between x and $j(x)$, where all slopes are 1. Now

$$g(\sigma) - \frac{L(L+1)}{2} = \sum_x x(\sigma_x - 1) = \sum_{\sigma_j = 2} j - \sum_{\sigma_x = 0} x = \sum_{\sigma_x = 0} \big(j(x) - x\big) + \sum_{\substack{\sigma_j = 2 \\ j \notin j(\cdot)}} j \;\geq 0.$$

If $\sigma_L = 2$, the site L contributes either L to the last sum or $L - x \geq 1$ to the previous one, so the difference is at least 1. □

Corollary 6.3. *$K(\rho) \leq L(L+1)/2$ for every recurrent ρ, with equality for $\rho = \mathbf{1}$.*

Proof. By Proposition 6.1 and Lemma 6.2, $K(\rho) = L(L+1) - g(\rho) + [\,\rho_L = 2\,] \leq L(L+1)/2$. For $\mathbf{1}$, $g = L(L+1)/2$ and $\rho_L = 1$. □

7. Proof of Corollary B

(1) If $n \geq L(L+1)$ then $n \geq K(\sigma)$ for every σ, because $K(\sigma) \leq L(L+1)$ with equality only when $g(\sigma) = 0$, i.e. $\sigma = 0$ (if $\sigma_L = 2$ then $g(\sigma) \geq 2L > 1$). So $\mathcal{W}^{n+1}|\sigma\rangle = |\Psi\rangle = \mathcal{W}^n|\sigma\rangle$ for all σ by Theorem A. For $n = L(L+1) - 1$, Theorem A gives $\mathcal{W}^n|0\rangle = v_u \neq |\Psi\rangle = \mathcal{W}^{n+1}|0\rangle$. For the minimal polynomial, $\mathcal{W}^{L(L+1)}(\mathcal{W} - 1) = 0$; the polynomial $x^{L(L+1)}$ does not annihilate $\mathcal{W}$ because $\mathcal{W}^{L(L+1)}|0\rangle = |\Psi\rangle \neq 0$, and $x^{L(L+1)-1}(x-1)$ does not by what we just proved; hence no $x^a(x-1)$ with $a < L(L+1)$ does. Every proper monic divisor of $x^{L(L+1)}(x-1)$ divides $x^{L(L+1)}$ or $x^{L(L+1)-1}(x-1)$, so the minimal polynomial is $x^{L(L+1)}(x-1)$, and the exponent of x is the size of the largest Jordan block for the eigenvalue 0.

(2) The span $\mathcal{V}_{\mathrm{rec}}$ is invariant under $\mathcal{W}$ by the closure property of Section 6. By Corollary 6.3 and Theorem A, $\mathcal{W}^{n+1} = \mathcal{W}^n$ on $\mathcal{V}_{\mathrm{rec}}$ for $n \geq L(L+1)/2$, while for $n = L(L+1)/2 - 1$ we have $\mathcal{W}^n|\mathbf{1}\rangle \neq |\Psi\rangle = \mathcal{W}^{n+1}|\mathbf{1}\rangle$. The minimal polynomial follows as in (1).

(3) Run the stationary chain, X_t with law Ψ. For observables X, Y on configurations and $\tau \geq L(L+1)/2$,

$$\mathbb{E}\big[X(X_t)Y(X_{t+\tau})\big] = \sum_{\rho} \Psi(\rho)X(\rho) \sum_{\rho'} (\mathcal{W}^{\tau}|\rho\rangle)(\rho')\, Y(\rho') = \sum_{\rho} \Psi(\rho)X(\rho) \sum_{\rho'} \Psi(\rho')Y(\rho'),$$

since only recurrent ρ contribute and $\mathcal{W}^{\tau}|\rho\rangle = |\Psi\rangle$ for them by Corollary 6.3; so $C_{XY}(\tau) = 0$. For $\tau_0 = L(L+1)/2 - 1$ take $X = [\,\cdot = \mathbf{1}\,]$ and $Y = [\,\cdot = \mathbf{2}\,]$. Starting from $\mathbf{1}$, after τ_0 grains the pile has at most $L(L+1)/2 + \tau_0 = L(L+1) - 1$ grains, so it is not in $\mathbf{2}$ (Lemma 4.3). Hence $C_{XY}(\tau_0) = 0 - \Psi(\mathbf{1})\Psi(\mathbf{2}) < 0$ by Proposition 6.1 and Lemma 4.3.

(4) If $\mathcal{C}$ has $m = g(\mathcal{C})$ grains, the exponent in (1) is $L(L+1) - m$, and the right side of (1) is $|\Psi\rangle$. By Theorem A the identity holds if and only if $L(L+1) - m \geq K(\mathcal{C})$, that is, if and only if $\mathcal{C}_L \neq 2$. The configuration $\mathbf{2}$ is recurrent (Lemma 4.3) and has $\mathbf{2}_L = 2$. □

Remark 7.1. The derivation in [5] of $\mathcal{W}^{L(L+1)/2}|\mathcal{R}\rangle = |\Psi\rangle$ for recurrent $\mathcal{R}$ passes through (1) and the lower bound $m \geq L(L+1)/2$. For a recurrent configuration with $\mathcal{R}_L = 2$ the identity (1) needs one extra grain, and the conclusion survives only because such configurations have at least $L(L+1)/2+1$ grains, which is Lemma 6.2. Dhar's argument for $\mathcal{W}^{L(L+1)+1} = \mathcal{W}^{L(L+1)}$, through $a_1^{L(L+1)} = a_1^2 \cdots a_L^2$, is correct and is recovered here as Corollary 3.5 with Corollary 3.3(1).

Declarations

Data availability. No data were generated. The LaTeX source of this paper and a short program that recomputes $\mathcal{W}$ exactly for $L \leq 6$ and checks Theorem A, Proposition 6.1 and Corollary B in exact rational arithmetic are archived at doi:10.5281/zenodo.24123896; no proof depends on the program.

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